INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – I, First Semester Examination, August – 2025

Subject Code: BCACC101

Subject Name: Introduction to C Programming

Full Marks: 30 Duration: 1 Hour

Part – A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. What is algorithm? Define flowchart. **2**
2. State the basic structure of a C program with an example. **2**
3. Analyse the functionality of **printf()** function. **2**
4. Can we have condition in **else**? **2**
5. Summarise the syntax of **for** loop with an example. **2**
6. Find the output of the following: **2**

```
void main()
{
   int year = 2000;
   if((year % 100)=100)
      printf("Probably leap year");
   else
      printf("Not a leap year");
}
```

Part – B

Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Create a C program to swap two integer values without using 3rd variable. **5**

8. Create a C program to read an integer value from the user. Then calculate the sum of digits of that number. **5**

9. In a table, describe all the built-in data types along with their name, keyword, size, and default value. **5**

Part – C

Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Design a C program to take total marks of a student & determine his/her division using following conditions:- **10**
if marks is >=600 then 1st divisions, or if marks <600 but marks>=450 then 2nd division, or marks<450 but marks>=300 then 3rd divisions. Otherwise fail.

11. In an examination, if a student score above 599 then his/her grade will be A. If marks is 599 or less, but more than 449 then grade will be B. If marks is 449 or less, but more than 299 grades will be C. Otherwise grade will be F. **10**

Develop an application using C to model above problem.

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